

The two-dimensional log n asymptotic as a genuine double integral

Claude, with Daniel Murfet

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Abstract

A short companion to unit 35: the two-dimensional standard integral is identified with the integral over the product set $(0, b]^2 \subseteq \mathbb{R}^2$ against two-dimensional Lebesgue measure, so that the first-log n theorem reads exactly in the paper's form

$$\int_{(0, b]^2} e^{-\beta n(uv)^2 + \beta \sqrt{n} uv a} d(u, v) \sim \frac{S_{1/2}(a)}{4} n^{-1/2} \log n.$$

Zero sorry/axiom.

1 Setting

Unit 35 proved the asymptotic for the iterated integral, deliberately avoiding Fubini. The paper's standard integral is $\int_{[0, b]^d}$; matching that form is a one-lemma Fubini step, cheap because the integrand is continuous and the chart is compact.

2 The things formalised

```
noncomputable def twoDIntegralProd ( a b n : ) : :=
  z in Ioc (0:) b × Ioc (0:) b,
  Real.exp ( - * n * (z.1 * z.2) ^ 2 + * Real.sqrt n * (z.1 * z.2) * a)

theorem twoDIntegralProd_eq ( a b n : ) : twoDIntegralProd a b n = twoDIntegral a b n
theorem twoDIntegralProd_isEquivalent ( a b : ) (h : 0 < ) (hb : 0 < b) :
  (fun n => twoDIntegralProd a b n) ~[atTop]
  fun n => fluctuation (1/2) a / 4 * (n ^ (-(1/2 : ))) * Real.log n
```

3 Proof strategy

Lebesgue measure on $\mathbb{R}^2$ is definitionally the product measure, and `Measure.prod_restrict` identifies the restriction to $S \times T$ with the product of restrictions; `integral_prod` then turns the double integral into the iterated one, given integrability on the product, which follows from continuity on the compact $[0, b]^2$ (`ContinuousOn.integrableOn_compact`, `IsCompact.prod`) and monotonicity of `IntegrableOn` in the set. The asymptotic is transported by `funext`.

4 Roadblocks and resolutions

None; first-try compile. Two idioms: rewrite the restricted measure into the product of restrictions *before* applying `integral_prod`, and check `volume = volume.prod volume` by `rfl` rather than searching for a lemma.

5 What was Mathlib, what was new

Mathlib: `integral_prod`, `Measure.prod_restrict`, `ContinuousOn.integrableOn_compact`, `IsCompact.prod`, `Set.prod_mono`. New: the paper-form statement of the two-dimensional $\log n$ asymptotic.

6 Lessons

Prove multivariate asymptotics in iterated form (where one-dimensional tools apply), and add the product-measure identification as a separate, trivial lemma afterwards.

7 Follow-ups

General (k_1, k_2) , (h_1, h_2) in two dimensions — multiplicity 2 exactly when $(h_1 + 1)/(2k_1) = (h_2 + 1)/(2k_2)$, and the generic (non-log) case otherwise — and the next-order term.